LE NGUYEN DUY LUAN

Mobile Developer | On-device AI / Computer Vision
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Portfolio | GitHub | LinkedIn

Professional Summary

Motivated and detail-oriented Mobile Developer specializing in Native Android (Kotlin, Jetpack Compose) and On-device AI/Computer Vision (TensorFlow Lite). Passionate about building performance-optimized, offline-first mobile applications with real-time ML capabilities. Possesses strong problem-solving skills, solid understanding of fullstack architectures, and a dedication to writing clean, maintainable code.

Education

Nha Trang University — Graduated: Mar 2026
Bachelor of Information Technology — *Nha Trang, Vietnam*

- **GPA**: 7.94/10 (3.25/4.0)
- **Graduation Project**: Lung X-ray Disease Classification using Segmentation-guided Masking — Score: 9.1/10
- **Relevant Coursework**: Data Structures & Algorithms, Mobile Device Programming, Software Architecture & Design, Image Processing, Artificial Intelligence

Technical Skills

Languages: Java, Kotlin, JavaScript, Python, SQL
Mobile: Android SDK, Jetpack Compose, CameraX, Room DB, WorkManager, Coroutines & Flow
AI / Computer Vision: TensorFlow Lite, MediaPipe, MLKit (Pose Estimation), YOLO, OpenCV, Gemini API
Web / Backend: Firebase (Auth, Firestore, Storage), Supabase, REST API design, OkHttp
Tools: Git/GitHub, Android Studio, Gradle, ADB (Android Debug Bridge), KSP, VS Code, Figma (basic)
AI-Assisted Dev: Google Antigravity, OpenAI Codex
Languages (Spoken): Vietnamese (native), English (technical reading/writing)

Projects

VocabLensAI — Native Android English Learning Platform | *Personal Project*

- **Tech Stack**: Kotlin, Jetpack Compose, Coroutines/Flow, Room DB, CameraX, TensorFlow Lite, Gemini API, Firebase, Supabase.
- Architected and developed an offline-first Android application from scratch using Kotlin, Jetpack Compose, MVVM, and clean code practices.
- Implemented real-time on-device object detection using CameraX and a lightweight TensorFlow Lite model, integrating Gemini AI to generate contextual learning stories and analyze pronunciation.
- Designed a custom learning engine incorporating the SM-2 Spaced Repetition algorithm for automated review scheduling, utilizing Room DB and Firestore for hybrid offline-first progress syncing.
- Built a real-time multiplayer PvP Arena using Firebase Firestore with an ELO rating calculator ($K = 40/20$), alongside a Chess.com-style game review UI for match accuracy analysis and AI feedback.
- Leveraged AI coding agents (Antigravity, Codex) to accelerate boilerplate generation and refactoring, while maintaining full manual review of architecture and core business logic.

Lung X-ray Disease Classification | *Graduation Project, Nha Trang University*

- **Tech Stack**: Python, TensorFlow/Keras, OpenCV, U-Net (Segmentation), ResNet/DenseNet (Classification).
- Developed a segmentation-guided masking pipeline to isolate lung regions in chest X-ray images before classification, improving model focus on diagnostically relevant areas and reducing noise.
- Trained, validated, and fine-tuned deep learning classification models, achieving a high degree of technical accuracy and a final project score of 9.1/10.
- Documented the end-to-end workflow from data preprocessing and image augmentation to model evaluation, creating a foundation suitable for medical-imaging research handoffs.

AI Fitness Coach (Android) | *Personal Project*

- **Tech Stack**: Kotlin, Jetpack Compose, MediaPipe/MLKit (Pose Estimation), Room DB, CameraX.
- Developed an offline-first Android application from scratch combining personalized workout planning, exercise tracking, and real-time form feedback.

- Implemented real-time camera-based pose estimation for squats, push-ups, and planks using MediaPipe/MLKit to extract coordinate joints and calculate movement angles.
- Built a custom rule-based analysis engine providing instant visual corrective feedback (e.g., ”Keep your back straight”) and automated rep counting.
- Designed local Room database caching for workout history, exercise metadata, and progress tracking, achieving zero-network dependency for core features.